
Supervised Learning

Model Evaluation Report

Linear Regression | Logistic Regression

Week 4 Internship Task

Prepared by

Thembinkosi Spogter Phama

AnalystLab Africa Internship

July 2026

1. Introduction

This report presents the model evaluation for Week 4 of the AnalystLab Africa Data Science Internship. The task focused on supervised learning - building predictive models using labeled datasets. Two models were developed: a Linear Regression model to predict house prices, and a Logistic Regression model to predict passenger survival on the Titanic.

Supervised learning is the foundation of predictive modeling in data science. It involves training algorithms on labeled data where the target outcome is known, then evaluating their ability to make accurate predictions on unseen data.

2. Datasets Used

Both datasets were cleaned and prepared during Weeks 1-2 of the internship.

2.1 Housing Price Dataset

Property	Value	Property	Value
Samples	545	Features	12
Target	Price	Task	Regression
Train Size	436 (80%)	Test Size	109 (20%)

2.2 Titanic Dataset

Property	Value	Property	Value
Samples	891	Features	7
Target	Survived	Task	Classification
Train Size	712 (80%)	Test Size	179 (20%)

3. Steps Taken

Step 1: Data Preparation

Categorical variables were encoded using Label Encoding. The Housing dataset had 7 categorical columns (yes/no) converted to binary. The Titanic dataset had Sex and Embarked encoded.

Step 2: Train/Test Split

Both datasets were split into training (80%) and testing (20%) sets using scikit-learn `train_test_split` with `random_state=42` for reproducibility.

Step 3: Model Training - Linear Regression

A Linear Regression model was trained on the Housing dataset to predict house prices. The model learns coefficients for each feature that minimize the mean squared error.

Step 4: Model Training - Logistic Regression

A Logistic Regression model was trained on the Titanic dataset to predict survival. The model estimates the probability of survival using the sigmoid function.

Step 5: Prediction & Evaluation

Both models made predictions on their respective test sets. Performance was evaluated using appropriate metrics: RMSE and R-squared for regression, Accuracy and Classification Report for classification.

4. Model Performance

4.1 Linear Regression - Housing Price Prediction

Metric	Value	Interpretation
R-squared	0.6495	Model explains 64.9% of price variance
RMSE	1,331,071	Avg prediction error of 1,331,071
RMSE/Mean	26.6%	Error relative to average price
Samples Tested	109	20% holdout set

The Linear Regression model performs moderately well with an R-squared of 0.65 (65%). This means approximately 65% of the variation in house prices can be explained by the features used. The average prediction error of 1,331,071 represents about 27% of the mean house price, indicating reasonable but not exceptional accuracy.

Top predictors: Bathrooms (+1,097,117), Air Conditioning (+785,551), Hot Water Heating (+687,881). Interestingly, Area has a surprisingly low coefficient, suggesting its effect may be captured by other correlated features.

4.2 Logistic Regression - Titanic Survival Prediction

Metric	Value	Interpretation
Accuracy	0.8101	Correctly classified 81.0% of passengers
Precision (Survived)	0.79	Of those predicted to survive, how many actually did
Recall (Survived)	0.74	Of actual survivors, how many were found
F1-Score (Survived)	0.76	Harmonic mean of precision and recall
Precision (Died)	0.83	Of those predicted to die, how many actually died
Recall (Died)	0.86	Of actual non-survivors, how many were found

The Logistic Regression model achieved 81.0% accuracy, correctly classifying 145 out of 179 test passengers. The model is better at identifying non-survivors (recall: 86%) than survivors (recall: 74%). This is likely due to the class imbalance in the dataset (more passengers died than survived).

Top predictors: Sex (dominant, coefficient -2.60), Pclass (-0.93), SibSp (-0.29). Gender is overwhelmingly the most important feature, consistent with the historical "women and children first" evacuation protocol.

5. Challenges Encountered

1. Feature Encoding

Categorical variables with multiple categories required careful encoding. Label Encoding was chosen for simplicity, but One-Hot Encoding may be explored in future weeks for potentially better performance.

2. Feature Scale Differences

Features have vastly different scales (e.g., Area in thousands vs binary yes/no columns). While Linear Regression handles this mathematically, it makes coefficient interpretation challenging. Standardization could improve interpretability.

3. Class Imbalance - Titanic

The Titanic dataset has more non-survivors (549) than survivors (342). The model reflects this bias, showing higher recall for the majority class. Techniques like class weights or SMOTE could address this.

4. Multicollinearity in Housing

Several features are correlated (e.g., bedrooms, bathrooms, area). This can affect coefficient stability and interpretation. Area showing a low coefficient may be due to its effect being split across correlated features.

5. Model Limitations

Linear Regression assumes a linear relationship and normally distributed residuals. The residual plot shows some patterns suggesting non-linear relationships exist. Advanced models in Week 5 may capture these better.

6. Overfitting Risk

With only 545 Housing samples and 12 features, there is a risk of overfitting. Cross-validation in future weeks will provide more robust performance estimates.

6. Key Insights

Linear Regression captures 65% of price variation

The R-squared of 0.65 indicates the model explains approximately 65% of the variance in house prices. This is a solid baseline, suggesting the features are informative but there is room for improvement with advanced models.

Bathrooms and AC are the strongest price drivers

Despite Area having the highest correlation with price (0.54), Bathrooms (+1,097,117) and Air Conditioning (+785,551) emerged as the strongest model coefficients. This suggests these features capture quality aspects beyond just size.

Logistic Regression achieves 81% on survival prediction

With 81% accuracy, the model provides a strong baseline for predicting passenger survival. Gender and class dominate the prediction, aligning with historical accounts of the disaster.

Gender is overwhelmingly the most important survival predictor

The Sex coefficient (-2.60) is nearly three times larger than the next feature (Pclass: -0.93). This confirms that being female was the single most important factor for survival, consistent with the "women and children first" protocol.

Model shows bias toward majority class

The recall for non-survivors (86%) exceeds recall for survivors (74%). This class imbalance effect should be addressed in future iterations through techniques like balanced class weights.

Simple models provide interpretable baselines

Both Linear and Logistic Regression offer clear, interpretable coefficients that explain exactly how each feature influences the prediction. This transparency is valuable before moving to more complex black-box models.

7. Conclusion

Week 4 successfully introduced supervised learning through two fundamental algorithms. The Linear Regression model explained 65% of housing price variance, while the Logistic Regression model achieved 81% accuracy on survival prediction. Both models serve as strong baselines for the advanced machine learning techniques to be explored in Week 5.

The key lesson learned is that simple, interpretable models are an essential first step in any machine learning project. They provide a performance benchmark and reveal which features are most influential before moving to more complex algorithms.

Prepared by

AnalystLab Africa Data Science Intern